

COLUMN | RAJEN VAGADIA

Smart will get smarter, faster, as we enter the future of wireless technologies



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Technology is evolving at such a rapid pace that it is very hard to predict where it will be a couple of decades from now. We can, however, make a guess at the general direction it will take. The immediate future, of course, is 5G, which, in conjunction with Artificial Intelligence (AI) and the Internet of Things (IoT), will enable “connected intelligence” – capabilities that will power industries and businesses and form the bedrock of most technology advancements of this decade. These technologies will enhance the way the world communicates, connects, and computes. What is different from the past, is the pace of the change, that these technologies will bring.

Equally important, they will have a significant positive impact on industries and economic activities,

which, in turn will boost national and global economies. A report by HIS Markit estimates that, between 2020 and 2035, global real GDP will grow at an average annual rate of 2.7%, of which 5G will contribute almost 0.2%. 5G could enable up to 5.9% of agricultural and 6.4% of public service (government) output in 2035. Moreover, the ability to create sustainable solutions using 5G, AI, IoT and other advanced technologies can impact important social goals and move the needle on some very important Sustainable Development Goals (SDGs), including those for good health and wellbeing; quality education; clean water and sanitation; decent work and economic growth; industry innovation and infrastructure; reduced inequalities; and sustainable cities and communities.

For now, the launch of 5G in India is imminent and eagerly awaited. The device ecosystem has been ready for over a year now; leading network operators have successfully demonstrated their readiness to roll out 5G; partnerships have been announced between telcos, OEMs and handset makers and most recently the government announced that operators can now start trials for 5G and its applications in various sectors. 5G will fuel the development of very unique business cases, across various verticals, industries, businesses and fields of which we today would not be able to fathom.

Connected Intelligence for smart home, smart cities, and smart lives

We are already seeing the emergence of devices that communicate with

each other and with us; they are learning from our behavior patterns and habits to simplify, or even automate, many routine activities. As the technology evolves further, we will see a greater progression, for instance, in the emergence of wearables that monitor and alert on personal health and well-being, anticipate or suggest our next move, and keep things in readiness for it. Imagine returning home after a jog and finding that the air conditioning has been turned on just before your arrival, set at the optimal level anticipating your core body temperature after the jog. Imagine wearables that monitor and note sudden changes in your core health parameters and automatically alert your emergency contacts and your doctor, thus ensuring that you get timely help. Or imagine being able to rush off for an appointment without worrying about securing your home, because you know your wearable will communicate with your smart home to I manage that. Home care has become a topic of great interest in these times and connectivity and AI are going to enable a lot, in realtime, of what would otherwise only be available through a hospital visit. This will save more lives, due to timely medical intervention.

The automotive industry will witness a huge transformation with the advent of connected intelligence. There will be a huge market for electric vehicles (EVs), autonomous driving, and personalized in-car experiences. As EVs become mainstream, we will see the deployment of technologies that enable them to charge wirelessly. EV batteries, which account for a significant